

SESSION 8 — Variable Expansion and Built-in Commands

Question 1

Demonstrate environment variable creation and variable expansion in Linux.

Commands

```
bhavan@LAPTOP-FGDPB110:~$ which gcc
/usr/bin/gcc
bhavan@LAPTOP-FGDPB110:~$ type ls
ls is aliased to 'ls --color=auto'
bhavan@LAPTOP-FGDPB110:~$ ls -l /bin/ls
-rwxr-xr-x 1 root root 142312 Jan 23  2026 /bin/ls
bhavan@LAPTOP-FGDPB110:~$ command -v gcc
/usr/bin/gcc
bhavan@LAPTOP-FGDPB110:~$ name="bhavan"
bhavan@LAPTOP-FGDPB110:~$ echo $name
bhavan
bhavan@LAPTOP-FGDPB110:~$ college="KL University"
bhavan@LAPTOP-FGDPB110:~$ echo $college
KL University
bhavan@LAPTOP-FGDPB110:~$ echo "My name is $name"
My name is bhavan
bhavan@LAPTOP-FGDPB110:~$ echo $unknown

bhavan@LAPTOP-FGDPB110:~$ student="$name"
bhavan@LAPTOP-FGDPB110:~$ echo $student
bhavan
bhavan@LAPTOP-FGDPB110:~$ |
```

Question 2

Demonstrate some built-in shell commands

```
bhavan@LAPTOP-FGDPB110:~$ pwd
/home/teja
bhavan@LAPTOP-FGDPB110:~$ cd /tmp
bhavan@LAPTOP-FGDPB110:/tmp$
bhavan@LAPTOP-FGDPB110:/tmp$ pwd
/tmp
bhavan@LAPTOP-FGDPB110:/tmp$ cd ~
bhavan@LAPTOP-FGDPB110:~$ echo "Operating Systems Lab"
Operating Systems Lab
bhavan@LAPTOP-FGDPB110:~$ type echo
echo is a shell builtin
bhavan@LAPTOP-FGDPB110:~$ |
```

Shell variables are used to store values. We use `$` before a variable name to access its value. Built-in commands such as `cd`, `pwd`, and `echo` are used directly by the shell to perform basic operations.